From Molecular Trajectories to Composable Paths

An exploratory categorical atlas of alanine dipeptide

Abstract. Can molecular geometry make the composition of dynamical behavior visible? We construct an interactive atlas from 750,000 publicly available molecular dynamics (MD) frames of alanine dipeptide. Molecular envelopes form a landscape over backbone-angle space. Two distinct categorical nerves can be superimposed: an overlap face-poset nerve and a free path-category nerve derived from observed basin passages. Translucent pathway skins now deform through an atom-attached displacement field, rather than merely changing illumination. This construction is a visual experiment and a precise vocabulary for discussing composition. It is not a claim of recovered configuration-space topology, a validated Markov model, or a completed theory of compositional protein dynamics. The prototype motivates an experimentally testable route from trajectory composition toward composition of open dynamical subsystems.

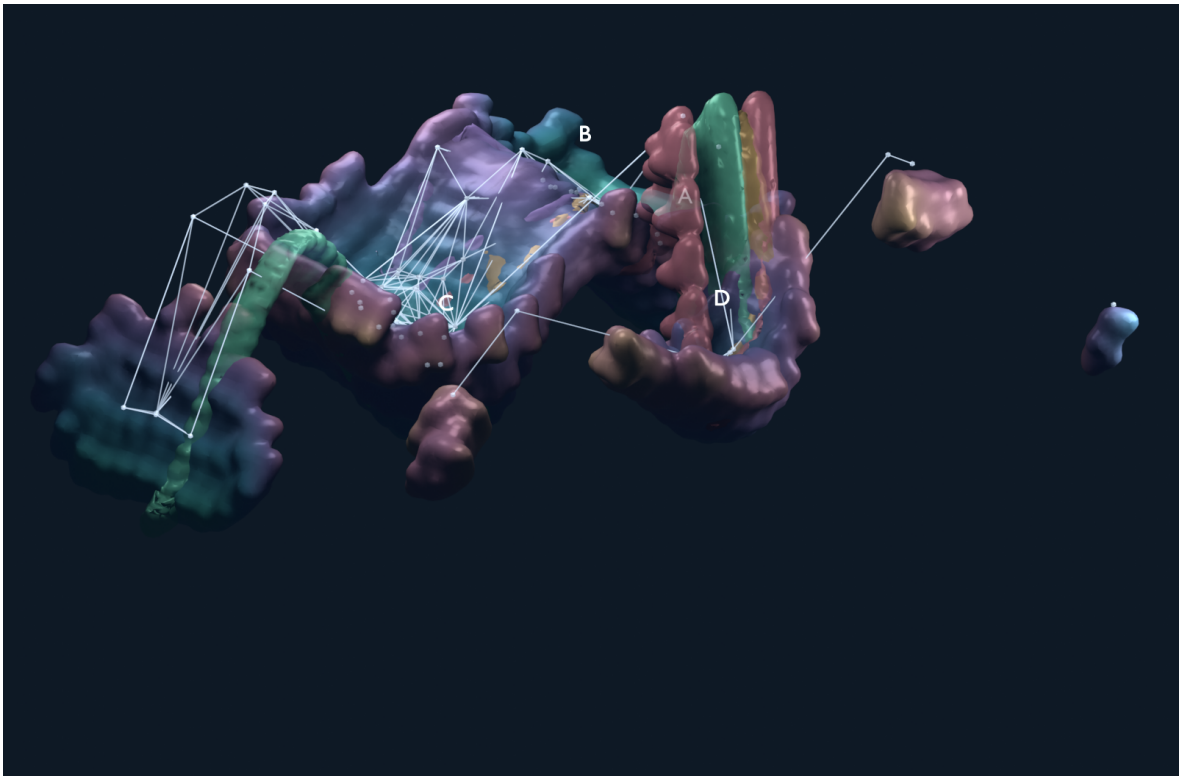


Figure 1: **A molecular landscape with a superimposed nerve.** Blender 4.0/Eevee rendering of the viewer's meshes and deformation targets. Terrain color follows empirical relative free energy (mint: populated; violet/amber: less populated). Pathway skins use source-basin colors. Silver nodes and edges show the overlap face-poset nerve. Labels A–D identify density modes. Horizontal coordinates encode ϕ, ψ; vertical placement encodes $1.3\,F/k_BT$ plus molecular relief. Scene distances are visual units.

Explore the artifact. Interactive molecular atlas (owner-private at preparation time). The viewer supports deformation amplitude, transparency, route filtering, scrubbing, and independent nerve overlays. All figures are generated from the numerical geometry; they are not AI-generated illustrations.

1. Two nerves, two meanings of composition

Let $q_t = (\phi_t, \psi_t) \in \mathbb{T}^2$ describe the backbone angles, and let $X_t \in \mathbb{R}^{10\times 3}$ contain the aligned heavy-atom coordinates. Alanine dipeptide is a capped amino-acid benchmark, not a folded protein. Its small conformational chart makes the distinctions below inspectable.

Overlap category. Take 18 balls of radius 73° in a cut angular chart. A finite sample W (every tenth MD frame) witnesses common membership:

$$K_W = \{\sigma \neq \varnothing : \exists q \in W,\ q \in \bigcap_{i\in\sigma} U_i\}.$$

We retain its 2-skeleton $K_W^{(2)}$ and form the face-poset category $\mathcal{C}_\cap = P(K_W^{(2)})$. Objects are simplices; arrows are inclusions of index sets, not reverse inclusions of intersection regions. Its nerve has 91 objects, 282 strict inclusions, and 204 nondegenerate 2-simplices. The realization is a barycentric subdivision:

$$|\mathrm{N}P(K_W^{(2)})| \cong |K_W^{(2)}|.$$

This categorical statement does not establish equivalence with the protein's configuration space. Good-cover hypotheses and sampling guarantees are separate issues [2, 3]. The overlap chart does not glue its periodic seam.

Dynamic path category. Let G have four basin objects and ten directed generators, one for each observed route type. Set $\mathcal{C}_{\mathrm{path}} = \mathrm{Free}(G)$. Morphisms are finite composable words; identities are empty words. For any category,

$$\mathrm{N}(\mathcal{C})_n = \mathrm{Fun}([n], \mathcal{C}) = \{x_0 \xrightarrow{f_1} \cdots \xrightarrow{f_n} x_n\}.$$

Inner face maps compose adjacent arrows; degeneracies insert identities. For a 2-simplex (f, g), $d_0(f, g) = g$, $d_1(f, g) = g \circ f$, and $d_2(f, g) = f$. Thus the long edge is a composite, not a missing observation to be filled in.

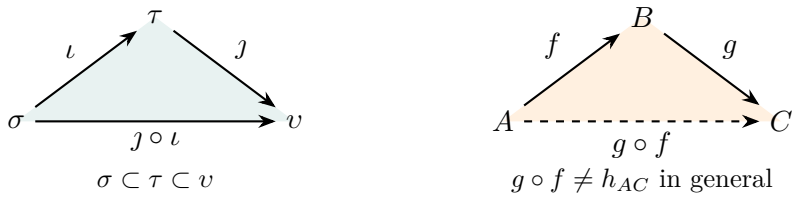


Figure 2: **The two displayed simplicial meanings.** Left: composition of inclusions. Right: formal composition of route generators. An independently observed $A \to C$ generator h_{AC} remains a different arrow. The displayed 25 pairs of composable generators are only a finite view of the infinite path-category nerve. Return paths are not identities.

The two categories coexist as overlays. No functor identifying them has been constructed. Likewise, a composable pair need not occur consecutively in the recorded MD trajectory.

2. Empirical dynamics before categorical interpretation

The MDshare benchmark contains three independent 250 ns trajectories at 300 K, with 1 ps frame spacing, generated using ACEMD, AMBER ff99SB-ILDN, and TIP3P water [1]. We use the supplied angles and aligned heavy-atom positions. All 750,000 frames contribute to occupancy estimates. Equal-area 12° angular bins are smoothed periodically with a Gaussian of width 0.85 bins:

$$\widehat{F}(q)/k_BT = -\log \frac{\widehat{p}(q)}{\max_{q'} \widehat{p}(q')}.$$

Pooling is unweighted. This is an empirical occupancy landscape; equilibrium convergence and estimator uncertainty have not been established.

Density modes are found separately on a 60×60 grid, smoothed at 1.4 bins. Disjoint 25° cores around four peaks define passage labels. The label ℓ_t retains the last core visited between core entries. Fifteen initial frames, before any core visit, are excluded from labeled counts.

Basin	ϕ	ψ	Last-core occupancy
A	−75°	153°	39.57%
B	−141°	159°	25.62%
C	−75°	−15°	31.82%
D	51°	33°	2.99%

These are density basins, not demonstrated metastable macrostates. In particular, A and B interconvert rapidly. Across independent runs, 73,145 core-to-core passages produce ten directed route types. Representative counts are A→B: 34,316; B→A: 34,323; A→C: 1,176; B→D: 1; and D→C: 2. Sparse directions remain visible and explicitly flagged.

Lagged probabilities. For lag $\tau \in \{10, 50, 100\}$ ps, count sliding pairs within each run:

$$C_{ij}(\tau) = \sum_r \sum_{t \in I_r(\tau)} \mathbf{1}\{\ell_t^{(r)} = i, \ell_{t+\tau}^{(r)} = j\}, \qquad \widehat{P}_{ij}(\tau) = \frac{C_{ij}(\tau)}{\sum_k C_{ik}(\tau)}.$$

Here $I_r(\tau)$ includes only valid labeled pairs in run r. Self transitions are included; pairs never cross run boundaries. Lagged probabilities may include indirect routes. Sliding observations are correlated, so row counts are not independent sample sizes. Gross passage frequency $n_{ij}/750\,\text{ns}$ is distinct from net current.

What has been checked. Independent counts obtained by removing non-core observations and counting changes reproduce the passage counts. Exported atomic frames match the raw coordinates after centering. The first 100 raw coordinate frames reproduce the supplied torsions to about 2.4×10^{-7} radians. Probability rows normalize; periodic-distance invariance, endpoint recovery and within-run lag accounting pass automated checks.

What remains unvalidated. There is no Chapman–Kolmogorov test, implied-timescale plateau, block-bootstrap uncertainty estimate, or detailed-balance assessment. The atlas should not be used to infer new rates, free-energy barriers or mechanistic causality.

3. From swept molecular skins to deforming geometry

For a representative passage, let $u \in [0,1]$ parameterize its recorded time. Coordinates are centered and scaled by 3.5 visual units per nm, and translated to $(\phi/12,\ 1.3\widehat{F}/k_BT,\ \psi/12)$. Interpolated atomic Gaussians of width 0.36 visual units form a swept envelope:

$$\rho_u(x) = \sum_a \exp\left[-\frac{\|x - Y_a(u)\|^2}{2(0.36)^2}\right], \qquad S = \{x : \max_u \rho_u(x) = 0.7\}.$$

The implementation uses at least 96 temporal samples, a 0.22-unit grid, and 0.45-voxel smoothing before isosurface extraction. Skins share the terrain's height and molecular scale; they are translucent intersecting meshes, not a remeshed Boolean union. Cut-chart crossings are not bridged.

Actual vertex motion. Each skin vertex v receives an approximate nearest-atom phase u_v and normalized atom weights $w_{va} \propto \exp[-\|v - Y_a(u_v)\|^2/(2(0.7)^2)]$. We introduce a bounded material time warp:

$$\tau_t(u) = u + 0.08\sin(2\pi(t-u))\sin(\pi u), \qquad D_v(t) = \sum_a w_{va}\left[Y_a(\tau_t(u_v)) - Y_a(u_v)\right].$$

It fixes the passage endpoints and remains within $[0,1]$. After two adjacency-averaging steps L, the rendered vertex is

$$v(t) = v + B_{1.5}\Big(L^2 D_v(t)\Big), \qquad B_c(z) = c\tanh(\|z\|/c)\frac{z}{\|z\|}, \quad B_c(0) = 0.$$

Endpoint-attached displacements are reset to zero. Twelve cyclic keyframes store both positions and recomputed normals; the browser interpolates between neighboring targets. A deformation-amplitude control scales the displacement down to zero for direct comparison with the fixed-envelope version.

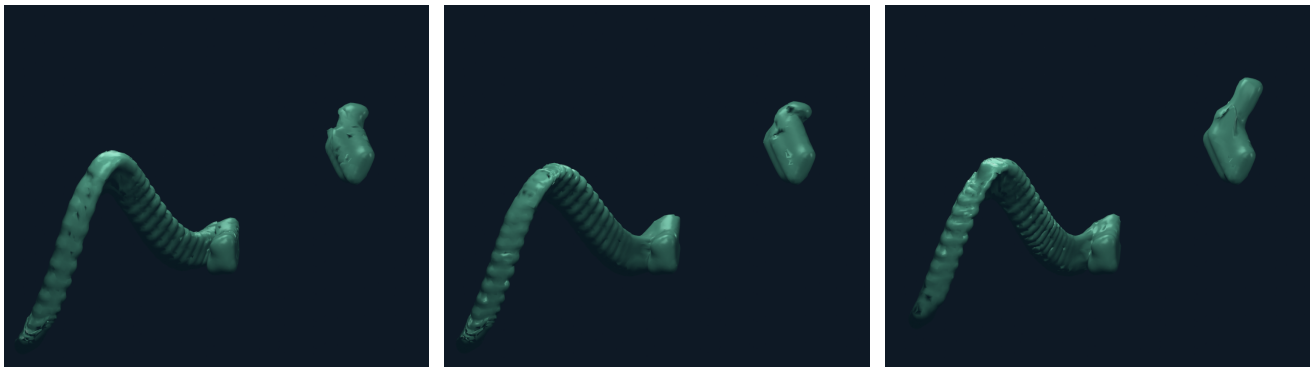


Figure 3: **Geometry changes, not only illumination.** Blender snapshots of the A→C skin at visual phases $t = 0, 1/3, 2/3$, with the same camera and lighting. Other route skins, terrain and the nerve are hidden. These phases are material deformation phases, not three complete instantaneous MD configurations.

This is an artistic, MD-attached deformation field. Interpolated Cartesian coordinates need not preserve bond lengths; self-overlaps can have ambiguous phase assignments. We do not assert injectivity, volume conservation, topology preservation, physical forces, or a dynamical semigroup for this animation.

4. A route toward compositional structural bioinformatics

The current construction makes *path composition* inspectable. Composing complete dynamical subsystems is a stronger objective. For an autonomous flow, one expects $\Phi_{t+s} = \Phi_t \circ \Phi_s$; a Markov approximation would analogously require $P(t+s) \approx P(t)P(s)$ under a stated convention. Neither property follows from drawing a nerve.

One proposed next step is to define structural interfaces (for example, selected contact or torsional observables), attach validated dynamics to modules, and specify how compatible interfaces compose. Open reaction networks and open Markov processes provide established categorical examples: input/output boundaries allow systems to compose, and functorial semantics can relate composite models to their behavior [4, 5]. Applying such ideas to structural bioinformatics would require new choices and validation, not just a change in visualization.

The research questions are concrete:

1. Can basin definitions and transition estimates remain stable across trajectory blocks, temperatures, and resolutions?
2. Can experimentally meaningful interfaces support compositional models with quantified approximation error?
3. Can a semantics functor $\mathcal{S} : \mathcal{D} \to \mathcal{B}$ satisfy $\mathcal{S}(g \circ f) = \mathcal{S}(g) \circ \mathcal{S}(f)$, where $\mathcal{D}$ is a category of specified open models and $\mathcal{B}$ their behaviors?

Here $\mathcal{D}$, $\mathcal{B}$, and $\mathcal{S}$ are research targets, not artifacts implemented by this prototype. Homotopy.io offers a useful reference for higher-categorical diagramming [6]; the current viewer is a custom Three.js implementation, without a homotopy.io export or proof certificate.

Reproducibility. The companion source includes data-download and analysis scripts, MD-driven morph targets, the LaTeX source, a Blender scene with shape keys, and the figure-generation script. The browser and Blender use the same geometry. The underlying source-file SHA-256 hashes are recorded in `dist/data.json`. Regenerate in this order: `download.py`, `prepare.py`, `dynamics.py`, `build_skins.py`, `build_deformation.py`. The figures use Blender 4.0/Eevee; the document is compiled from LaTeX.

Exploratory proposition. Categorical language may help separate state, transition, composition, and interface in molecular modeling. This atlas offers a visible, inspectable starting point for that discussion, rather than evidence that categorification has already improved a predictive model.

References

[1] MDshare. *Alanine dipeptide dataset and simulation metadata.* https://markovmodel.github.io/mdshare/ALA2/.

[2] J. P. May. *Simplicial sets, nerves of categories, Kan complexes, etc.* https://math.uchicago.edu/~may/REU2018/sset.pdf.

[3] U. Bauer, M. Kerber, F. Roll, A. Rolle. *A Unified View on the Functorial Nerve Theorem and its Variations.* https://arxiv.org/abs/2203.03571.

[4] J. C. Baez, B. S. Pollard. *A Compositional Framework for Reaction Networks* (2017). https://arxiv.org/abs/1704.02051.

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[6] N. Corbyn et al. *homotopy.io: a proof assistant for finitely-presented globular n-categories* (2024). https://arxiv.org/abs/2402.13179.